

Jules Berman

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Summary

Graduating PhD candidate building probabilistic and generative models for data-driven scientific simulation. Current research focuses on developing novel flow- and diffusion-based algorithms for applications in stochastic dynamical/physical systems. Author on **5 NeurIPS/ICML/ICLR papers** and **Harold Grad Prize winner** (top PhD '24).

Education

Ph.D., Computer Science — New York University, Courant Institute. GPA 3.9. expected June 2026

Awards: Harold Grad Prize (top PhD student, 2024) · MacCracken Fellowship (full award)

Advisor: Benjamin Peherstorfer

B.S., Computer Science — New York University. June 2017

Awards: cum laude

Select Publications

Leveraging Gauge Freedom for Learning Non-Gradient Population Dynamics [...] **in review** ▪ **ICML 2026**
[arxiv] **J. Berman**, T. Blickhan, B. Peherstorfer.

Hankel Singular Value Regularization for Highly Compressible State Space Models **NeurIPS 2025**
[arxiv] P. Schwerdtner, **J. Berman**, B. Peherstorfer.

Parametric model reduction of [...] stochastic systems via higher-order action matching **NeurIPS 2024**
[arxiv] **J. Berman**, T. Blickhan, B. Peherstorfer.

CoLoRA: Continuous low-rank adaptation for reduced neural modeling [...] **ICML 2024**
[arxiv] **J. Berman**, B. Peherstorfer.

Randomized Sparse Neural Galerkin Schemes with Deep Networks **spotlight** ▪ **NeurIPS 2023**
[arxiv] **J. Berman**, B. Peherstorfer.

Experience

Google May 2025 – Aug 2025

Research Engineer Intern

Developed approach to generating time series embeddings via masked autoregression using large scale transformers; cut downstream model size 3–5× while improving accuracy by 10–25% across datasets.

Meta May 2024 – Aug 2024

Research Scientist Intern, CTRL-Labs

Designed and trained diffusion-based generative models of EMG data, improving downstream gesture classification accuracy by 10% through synthetic data augmentation.

Flatiron Institute May 2021 – Aug 2022

Research Analyst, Center for Computational Neuroscience

Architected a distributed training platform for large-scale experiments (100k+ VAEs) and developed novel representational metrics enabling in-depth comparative analyses.

Bloomberg LP April 2018 – April 2021

Senior Software Engineer, Global Infrastructure Team

Deployed ML-based predictive systems to forecast future infrastructure demands from historical data.

Software: Python · JAX · PyTorch · Optax · Flax · NumPy · SLURM · Git · LaTeX

Talks: CSE24, CSE25, MoRE25, UQ26, Gen-modeling Study Group

Projects: Developed an online free-to-play poker web application [justpoker.games]